Matrices - Exercises Solutions

Exercise 9.1. *Evaluate the following:*

(i) $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$

(ii) $\begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix} - \begin{pmatrix} 7 \\ -11 \\ -13 \end{pmatrix}$

(iii) $\begin{pmatrix} 1 & 3 & 5 \\ 7 & 9 & 11 \end{pmatrix} + \begin{pmatrix} 2 & 3 \\ 5 & 7 \end{pmatrix}$

(i) $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 1+5 & 2+6 \\ 3+7 & 4+8 \end{pmatrix} = \begin{pmatrix} 6 & 8 \\ 10 & 12 \end{pmatrix}$

(ii) $\begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix} - \begin{pmatrix} 7 \\ -11 \\ -13 \end{pmatrix} = \begin{pmatrix} 2-7 \\ 3+11 \\ 5+13 \end{pmatrix} = \begin{pmatrix} -5 \\ 14 \\ 18 \end{pmatrix}$

(iii) Undefined since the left matrix is 2×3 and the right matrix is 2×2

Exercise 9.2. *Evaluate the following:*

(i) $2\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

(ii) $\frac{1}{2}\begin{pmatrix} 0 & -1 \\ 3 & 2 \\ 4 & -2 \end{pmatrix}$

(iii) $a\begin{pmatrix} 1 & 6 & 4 \\ 0 & 3 & -1 \end{pmatrix}$

(iv) $101\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} - 99\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$

(i) $2\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 4 \\ 6 & 8 \end{pmatrix}$

Matrix icon by Agan24 from the Noun Project

(ii) $\frac{1}{2}\begin{pmatrix} 0 & -1 \\ 3 & 2 \\ 4 & -2 \end{pmatrix} = \begin{pmatrix} 0 & -1/2 \\ 3/2 & 1 \\ 2 & -1 \end{pmatrix}$

(iii) $a\begin{pmatrix} 1 & 6 & 4 \\ 0 & 3 & -1 \end{pmatrix} = \begin{pmatrix} a & 6a & 4a \\ 0 & 3a & -a \end{pmatrix}$

(iv) $101\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} - 99\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = (101-99)\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 4 \\ 0 & 2 \end{pmatrix}$

Exercise 9.3. *Evaluate the following:*

(i) $\begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}^{\mathsf{T}}$

(ii) $\begin{pmatrix} 1 & 0 & -2 \\ 3 & -4 & 1 \end{pmatrix}^{\mathsf{T}}$

(iii) $\begin{pmatrix} 2 & 3 & 5 \end{pmatrix}^{\mathsf{T}}$

(i) $\begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}^{\mathsf{T}} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

(ii) $\begin{pmatrix} 1 & 0 & -2 \\ 3 & -4 & 1 \end{pmatrix}^{\mathsf{T}} = \begin{pmatrix} 1 & 3 \\ 0 & -4 \\ -2 & 1 \end{pmatrix}$

(iii) $\begin{pmatrix} 2 & 3 & 5 \end{pmatrix}^{\mathsf{T}} = \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix}$

Exercise 9.4. *Given the matrices*

$$A = \begin{pmatrix} 1 & 0 \\ -2 & 3 \end{pmatrix} \qquad B = \begin{pmatrix} 2 & 3 \\ 1 & 5 \end{pmatrix} \qquad C = \begin{pmatrix} 1 & 1 & 0 \\ 3 & -2 & 1 \end{pmatrix} \qquad D = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

calculate the following (where possible):

(i) AB;
(ii) BC;
(iii) CD;
(iv) CC^{T}

(i)

$$AB = \begin{pmatrix} 1 & 0 \\ -2 & 3 \end{pmatrix} \begin{pmatrix} 2 & 3 \\ 1 & 5 \end{pmatrix} = \begin{pmatrix} 1\cdot 2 + 0\cdot 1 & 1\cdot 3 + 0\cdot 5 \\ -2\cdot 2 + 3\cdot 1 & -2\cdot 3 + 3\cdot 5 \end{pmatrix} = \begin{pmatrix} 2+0 & 3+0 \\ -4+3 & -6+15 \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ -1 & 9 \end{pmatrix}$$

(ii)

$$BC = \begin{pmatrix} 2 & 3 \\ 1 & 5 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 3 & -2 & 1 \end{pmatrix} = \begin{pmatrix} 2\cdot 1 + 3\cdot 3 & 2\cdot 1 + 3\cdot -2 & 2\cdot 0 + 3\cdot 1 \\ 1\cdot 1 + 5\cdot 3 & 1\cdot 1 + 5\cdot -2 & 1\cdot 0 + 5\cdot 1 \end{pmatrix}$$
$$= \begin{pmatrix} 2+9 & 2-6 & 0+3 \\ 1+15 & 1-10 & 0+5 \end{pmatrix} = \begin{pmatrix} 11 & -4 & 3 \\ 16 & -9 & 5 \end{pmatrix}$$

(iii) CD is undefined since C has 3 columns and D only has 2 rows.

(iv)

$$CC^{\mathsf{T}} = \begin{pmatrix} 1 & 1 & 0 \\ 3 & -2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 1 & -2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1\cdot 1 + 1\cdot 1 + 0\cdot 0 & 1\cdot 3 + 1\cdot -2 + 0\cdot 1 \\ 3\cdot 1 + -2\cdot 1 + 1\cdot 0 & 3\cdot 3 + -2\cdot -2 + 1\cdot 1 \end{pmatrix}$$
$$= \begin{pmatrix} 1+1+0 & 3-2+0 \\ 3-2+0 & 9+4+1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 14 \end{pmatrix}$$

Exercise 9.5. *Calculate the following determinants:*

(i) $\begin{vmatrix} 5 & 2 \\ 3 & 4 \end{vmatrix}$

(ii) $\det \begin{pmatrix} a & b \\ ka & kb \end{pmatrix}$

(iii) $\begin{vmatrix} 2-\lambda & 3 \\ 5 & 6-\lambda \end{vmatrix}$

(i) $\begin{vmatrix} 5 & 2 \\ 3 & 4 \end{vmatrix} = 5\cdot 4 - 2\cdot 3 = 20 - 6 = 14$

(ii) $\det \begin{pmatrix} a & b \\ ka & kb \end{pmatrix} = akb - akb = 0$

(iii) $\begin{vmatrix} 2-\lambda & 3 \\ 5 & 6-\lambda \end{vmatrix} = (2-\lambda)(6-\lambda) - 3\cdot 5 = \lambda^2 - 8\lambda - 3$

Exercise 9.6. *Given the matrix*

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

calculate:

(i) M_{11};
(ii) M_{12};
(iii) M_{13}

(i) $M_{11} = \begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} = 5(9) - 6(8) = 45 - 48 = -3$

(ii) $M_{12} = \begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} = 4(9) - 6(7) = 36 - 42 = -6$

(iii) $M_{13} = \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix} = 4(8) - 5(7) = 32 - 35 = -3$

Exercise 9.7. *(a) Calculate the determinant of the matrix*

$$\begin{pmatrix} 1 & 0 & 4 \\ 2 & 5 & 6 \\ 4 & 5 & 2 \end{pmatrix},$$

by expanding along:
(i) row 1;
(ii) column 3

(i)

$$\begin{vmatrix} 1 & 0 & 4 \\ 2 & 5 & 6 \\ 4 & 5 & 2 \end{vmatrix} = 1\begin{vmatrix} 5 & 6 \\ 5 & 2 \end{vmatrix} - 0\begin{vmatrix} 2 & 6 \\ 4 & 2 \end{vmatrix} + 4\begin{vmatrix} 2 & 5 \\ 4 & 5 \end{vmatrix}$$
$$= (10 - 30) + 4(10 - 20) = -60$$

(ii)

$$\begin{vmatrix} 1 & 0 & 4 \\ 2 & 5 & 6 \\ 4 & 5 & 2 \end{vmatrix} = 4\begin{vmatrix} 2 & 5 \\ 4 & 5 \end{vmatrix} - 6\begin{vmatrix} 1 & 0 \\ 4 & 5 \end{vmatrix} + 2\begin{vmatrix} 1 & 0 \\ 2 & 5 \end{vmatrix}$$
$$= 4(10 - 20) - 6(5 - 0) + 2(5 - 0) = -60$$

(b) Calculate the determinant of the 4×4 *matrix*

$$\begin{pmatrix} 1 & -1 & 4 & 3 \\ 2 & 0 & 5 & -3 \\ 1 & 2 & 4 & 5 \\ 2 & 0 & -2 & 4 \end{pmatrix}$$

Here column 2 has two zero elements so it would be more efficient to expand along this column.

$$\begin{vmatrix} 1 & -1 & 4 & 3 \\ 2 & 0 & 5 & -3 \\ 1 & 2 & 4 & 5 \\ 2 & 0 & -2 & 4 \end{vmatrix} = -(-1)\begin{vmatrix} 2 & 5 & -3 \\ 1 & 4 & 5 \\ 2 & -2 & 4 \end{vmatrix} - 2\begin{vmatrix} 1 & 4 & 3 \\ 2 & 5 & -3 \\ 2 & -2 & 4 \end{vmatrix}$$

$$= 1\left(2\begin{vmatrix} 4 & 5 \\ -2 & 4 \end{vmatrix} - 5\begin{vmatrix} 1 & 5 \\ 2 & 4 \end{vmatrix} - 3\begin{vmatrix} 1 & 4 \\ 2 & -2 \end{vmatrix}\right) - 2\left(\begin{vmatrix} 5 & -3 \\ -2 & 4 \end{vmatrix} - 4\begin{vmatrix} 2 & -3 \\ 2 & 4 \end{vmatrix} + 3\begin{vmatrix} 2 & 5 \\ 2 & -2 \end{vmatrix}\right)$$

$$= 2(16+10) - 5(4-10) - 3(-2-8) - 2(20-6) + 8(8+6) - 6(-4-10)$$
$$= 280.$$